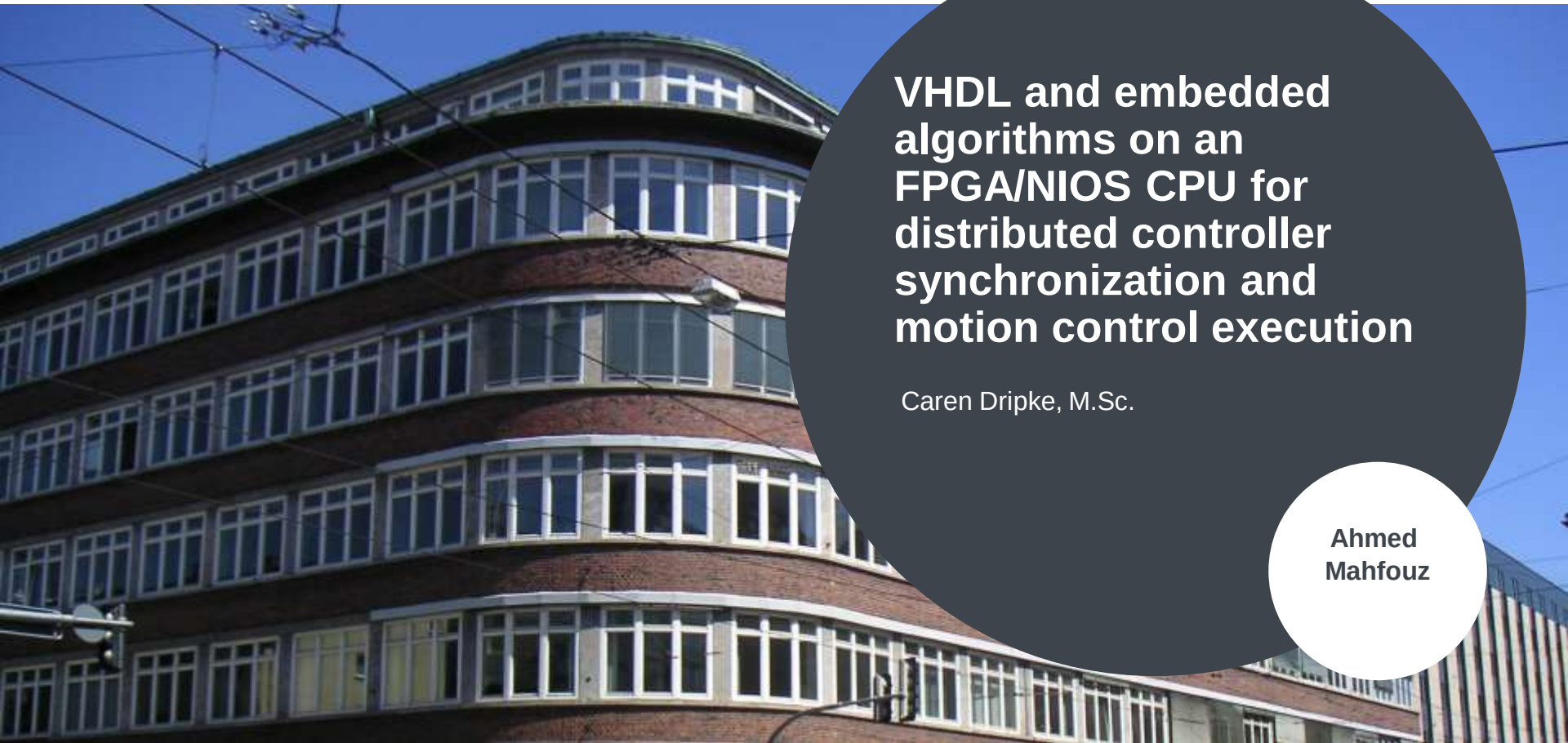
University of Stuttgart
Institute for Control Engineering of Machine Tools and Manufacturing Units (ISW)



VHDL and embedded algorithms on an FPGA/NIOS CPU for distributed controller synchronization and motion control execution

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About myself

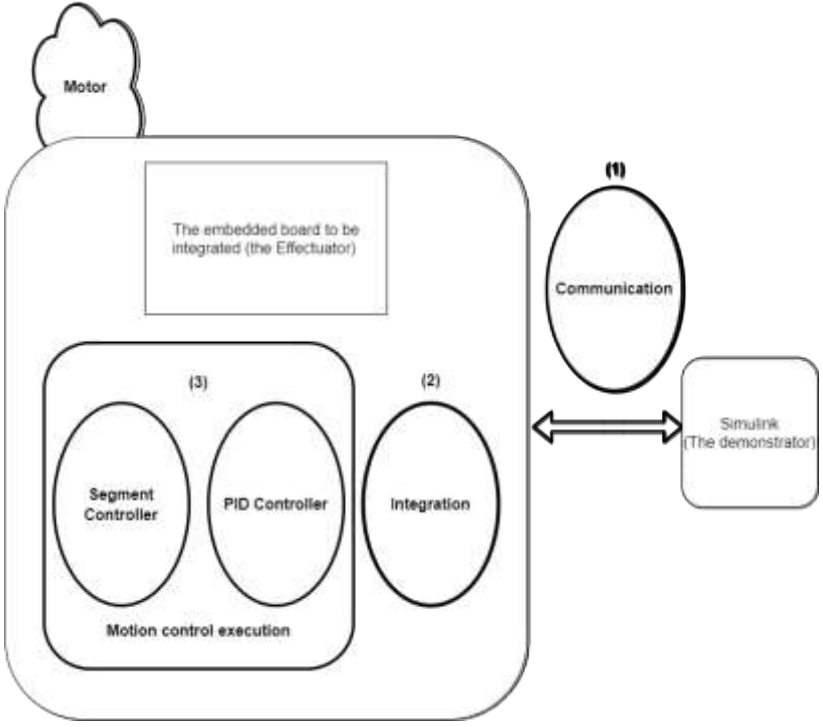
- Name: Ahmed Mahfouz
- Nationality: Egyptian
- Home city: Cairo, Egypt
- University: German University in Cairo (GUC)
- Faculty: Engineering and Material Science (EMS)
- Studies: Mechatronics Engineering bachelor student

About DEVEKOS research project [1]

- Definition
- Started officially on: 1st of March 2017 (still on work)
- Funded by: the Federal Ministry for Economic Affairs and Energy (BMWi)

Presentation outline

- ✓ Introduction
- ✓ Communication implementation
- ✓ The embedded board integration into the project setup
- ✓ Motion control execution
- ✓ Conclusion and results
- ✓ Further work recommendations

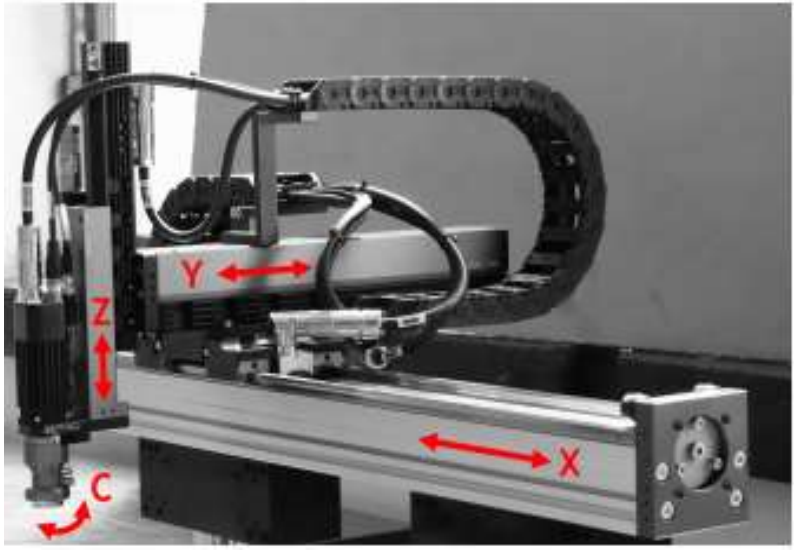


Presentation outline

Introduction

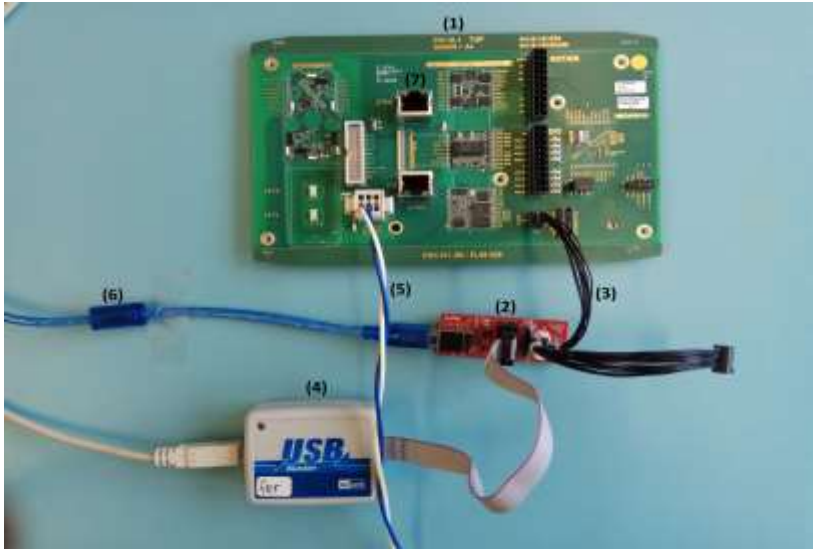
Initial setup of my thesis project

- The demonstrator existing at ISW
- A Simulink model running on the demonstrator



The demonstrator existing at ISW [2]

- An embedded NIOS CPU (the Effectuator)
- NIOS II C project

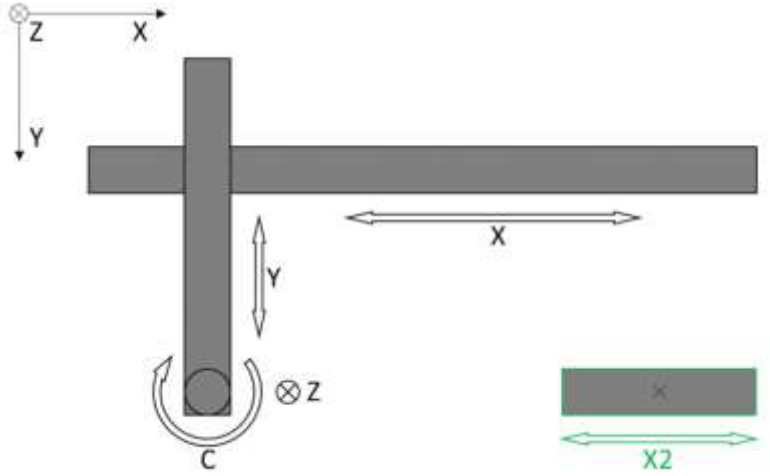


NIOS CPU board (the Effectuator) [3]

Introduction

The objective of my thesis

- Implementing a communication network between the demonstrator and the Effectuator
- Integration the Effectuator with the three transitional axes of the demonstrator
- Motion control execution for the Effectuator axis (X2-axis)



The Effectuator integrated with the demonstrator[4]

Communication implementation

The data shared between the Effectuator and the demonstrator

Path preparation calculations

- Coordination vector:
  - Axis 1 = [t1.1, t1.2, t1.3, ….., t1.n−1] , where n is the number of points in the axis path
- Coordination matrix:
  - Axis 1 = [t1.1, t1.2, t1.3, ….., t1.n−1]
  - Axis 2 = [t2.1, t2.2, t2.3, ….., t2.n−1]
  - Axis 3 = [t3.1, t3.2, t3.3, ….., t3.n−1]
  - Axis 4 = [t4.1, t4.2, t4.3, ….., t4.n−1]
  - .
  - Axis a = [ta.1, ta.2, ta.3, ….., ta.n−1], where a is the number of the system axes
- Consensus vector:
  - tmax.1= max(t1.1, t2.1, t3.1, t4.1, … ta.1)
  - tmax.2= max(t1.2, t2.2, t3.2, t4.2, … ta.2)
  - tmax.3= max(t1.3, t2.3, t3.3, t4.3, … ta.3)

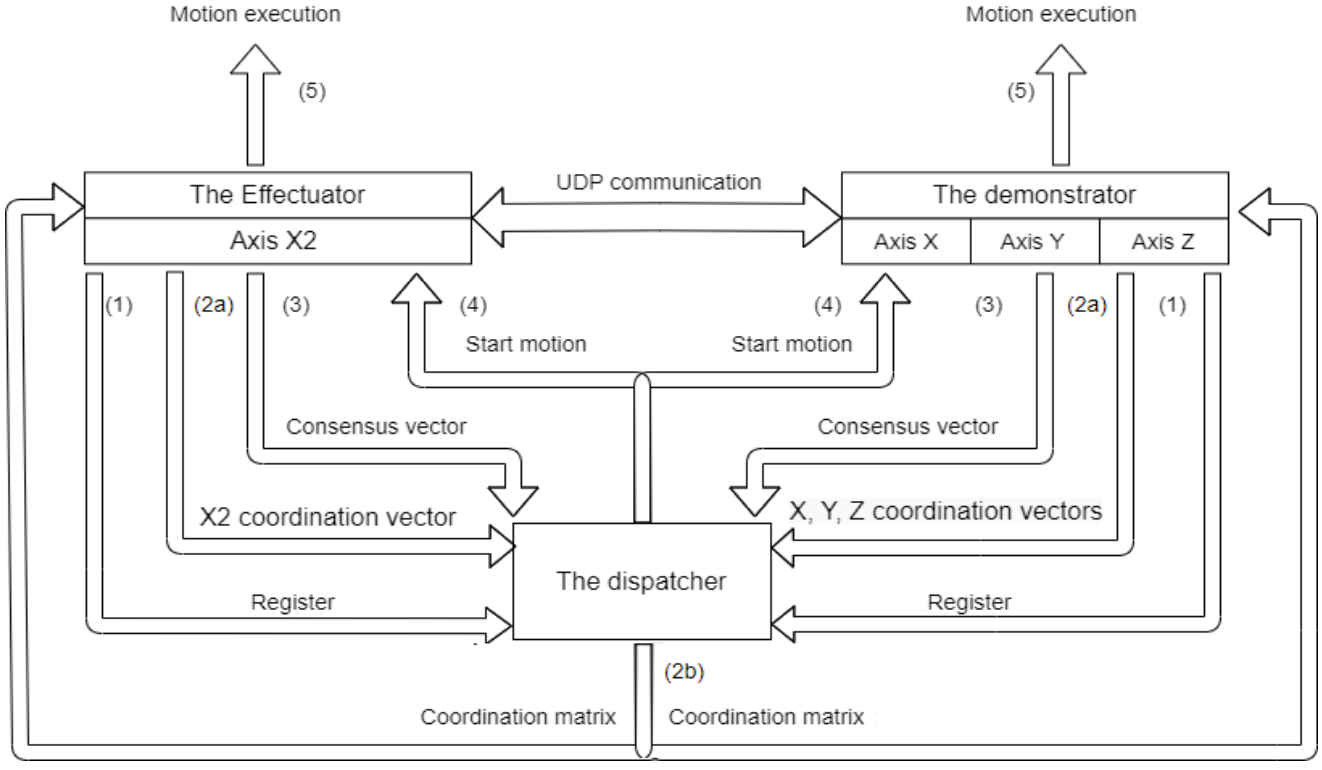
  Consens vector= [tmax.1, tmax.2 …., tmax.n-1]
- Execution vector:

  ExeVec =

  [path points; speeds; times; cummulatve times]

Communication implementation

The data communication sequence

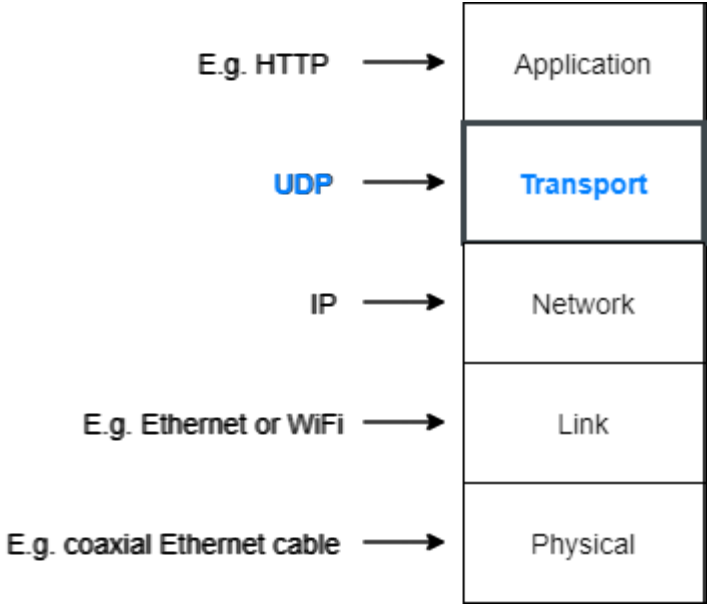


The entire process of the new system [4]

UDP communication implementation

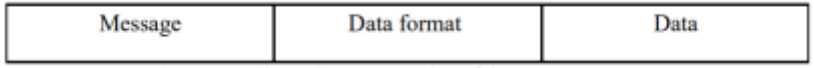
Communication protocol and Message structure

- What is UDP/IP (User Datagram Protocol) ?



Packet structure [5]

- Message structure



Schematic of message structure [4]

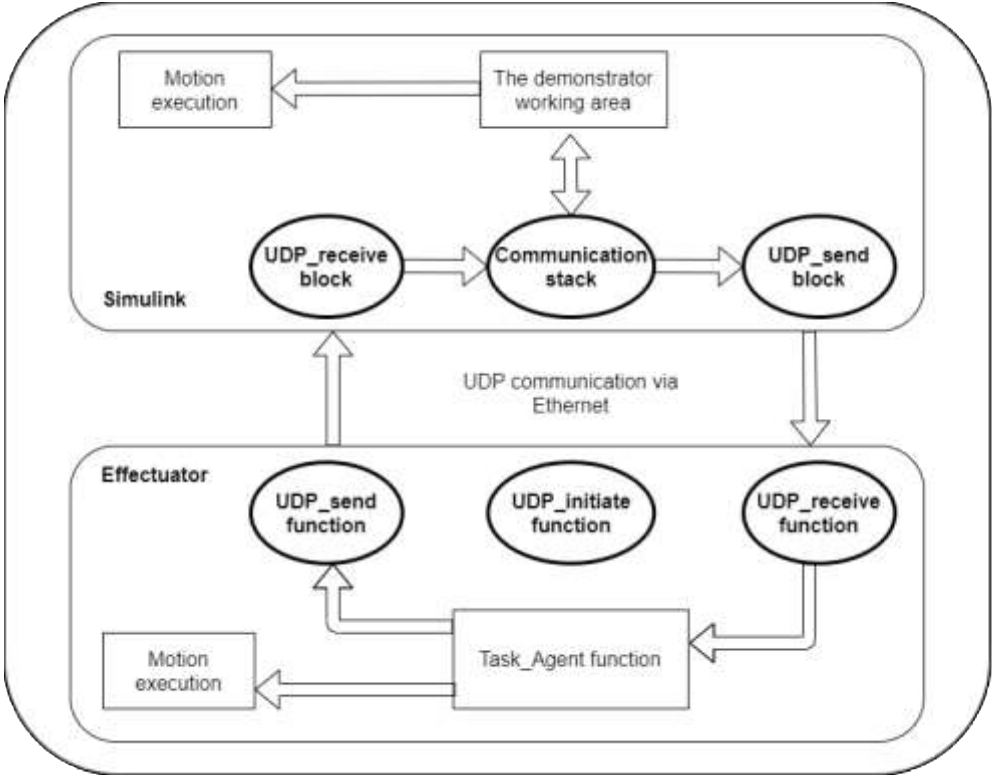
$$A = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \\ \vdots & \vdots & & \vdots \\ z_1 & z_2 & \dots & z_n \end{pmatrix}$$

Representation of data matrix [4]

Schematic of the data field [4]

UDP communication implementation

- Effectuator implmentation
  - UDP_initiate function
  - UDP_send function
  - UDP_receive function
- Simulink implementation
  - UDP_send block
  - UDP_receive block
  - Communication stack

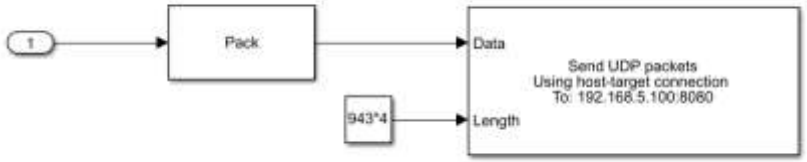


Communication Software structure

UDP communication implementation

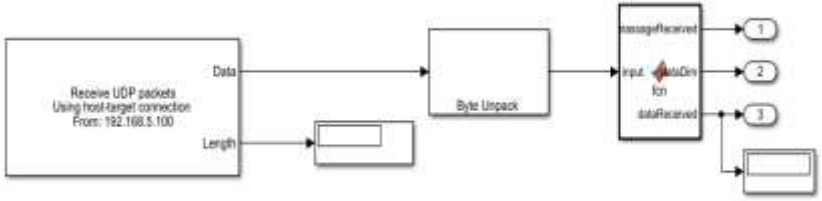
The Simulink implementation

- UDP_communication Simulink block
  - UDP_send block



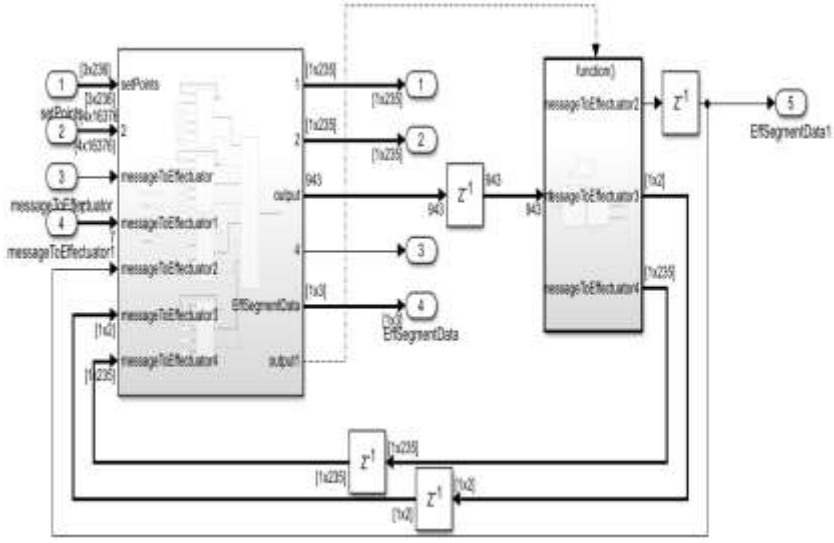
UDP_send block [4]

  - UDP_receive block



UDP_receive block [4]

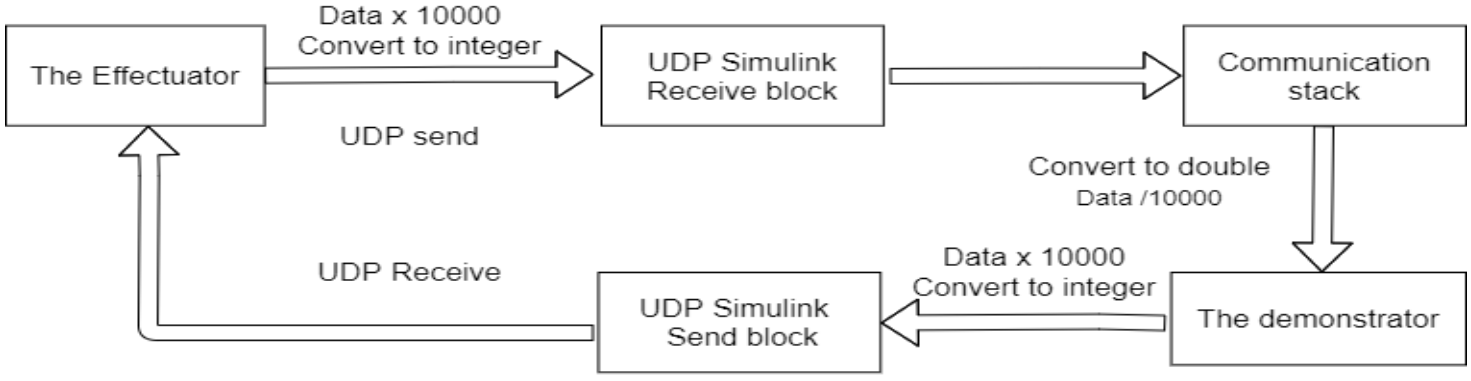
- Communication stack



Communication stack [4]

UDP communication implementation

Data type conversion

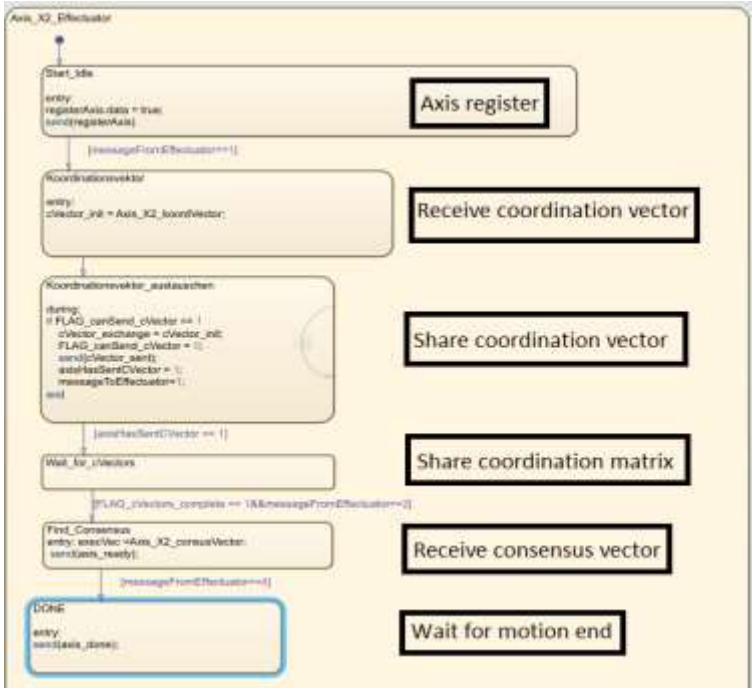


UDP communication process diagram [4]

X2-axis integration into the project setup

Inserting X2-axis into the demonstrator common working area

- The effectuator task_agent
  - Generate X2 curve
  - Generate coordination vector
  - Generate consensus vector
  - Generate execution vector
  - Execute movement
- The effectuator task_agent
  - X2-axis or the Effectuator flowchart



X2-axis flowchart [4]

X2-axis motion control implementation

PID controller embedded implementation

- Converting the transfer function of the PID into a difference equation

$$PID(z) = P + I.T_s \cdot \frac{1}{z-1} + D \cdot \frac{N}{1 + N \cdot T_s \cdot \frac{1}{z-1}} = \frac{U(z)}{E(z)} \quad (5.4)$$

Discrete PID controller transfer function [4]

Where **P** is the proportional gain, *I* is the integral gain, **D** is the derivative gain, Ts is the sampling time, **N** is the filter coefficient, **U(z)** is the transfer function output, **E(z)** is the transfer function input

X2-axis motion control implementation

PID controller embedded implementation

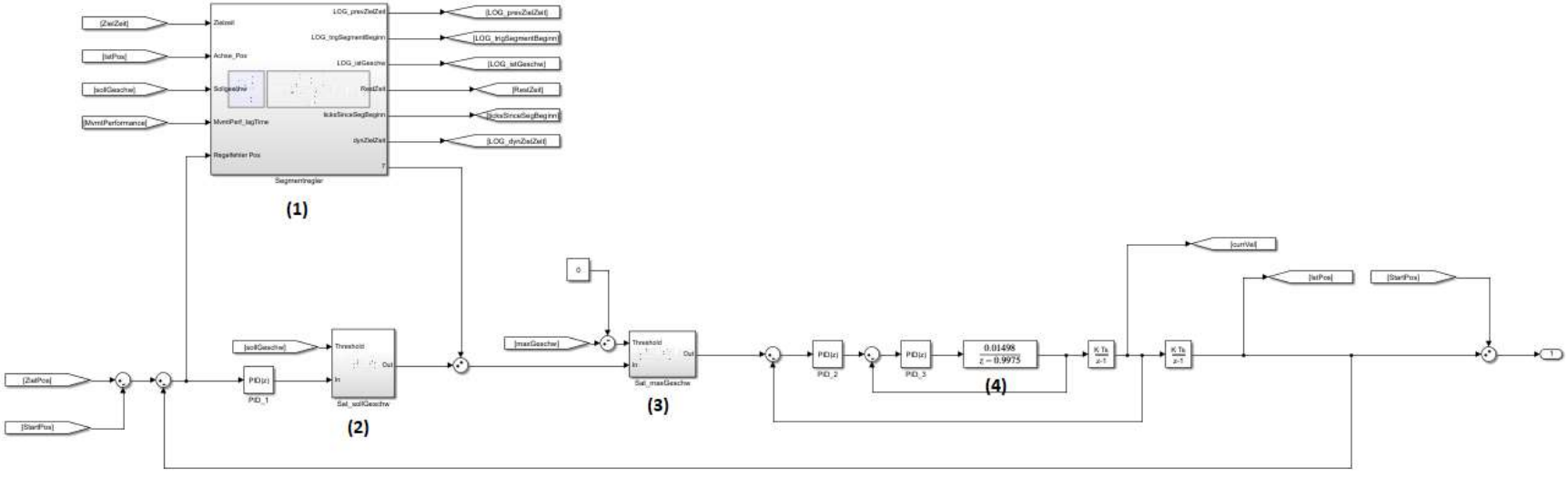
$$u(k) = (2 - N \cdot T_s)u(k-1) + (N \cdot T_s - 1)u(k-2) + (P + D \cdot N)e(k) + (P \cdot N \cdot T_s + I \cdot T_s - 2P - 2D \cdot N)e(k-1) + (P + I \cdot N \cdot T_s^2 - P \cdot N \cdot T_s - I \cdot T_s + D \cdot N)e(k-2) \quad (5.5)$$

Discrete PID controller a difference equation [4]

Where **u(k)** is the current PID output , **u(k-1)** is the previous PID output, **u(k-2)** is the former PID input, **e(k)** is the current PID output , **e(k-1)** is the previous PID input , **e(k-2)** is the former PID input

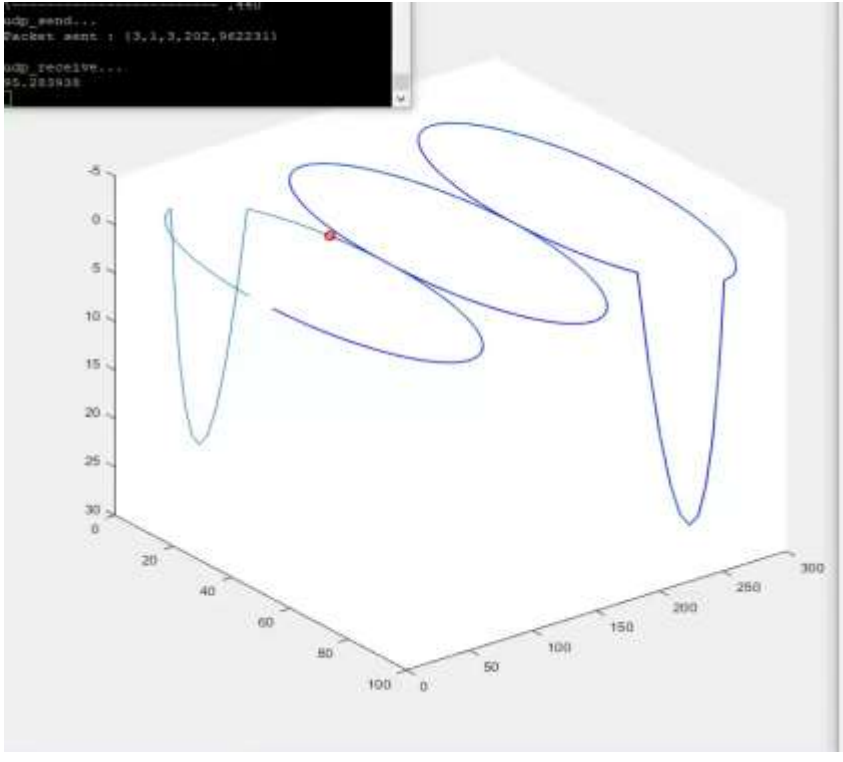
X2-axis motion control implmentation

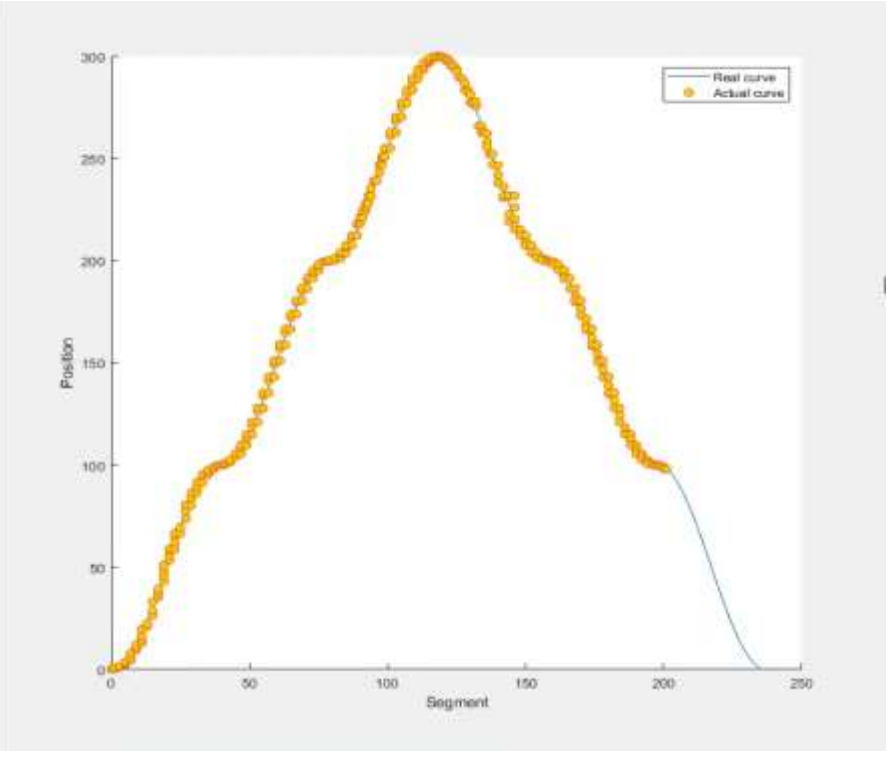
Axis control loop implementation



X-axis control loop [4]

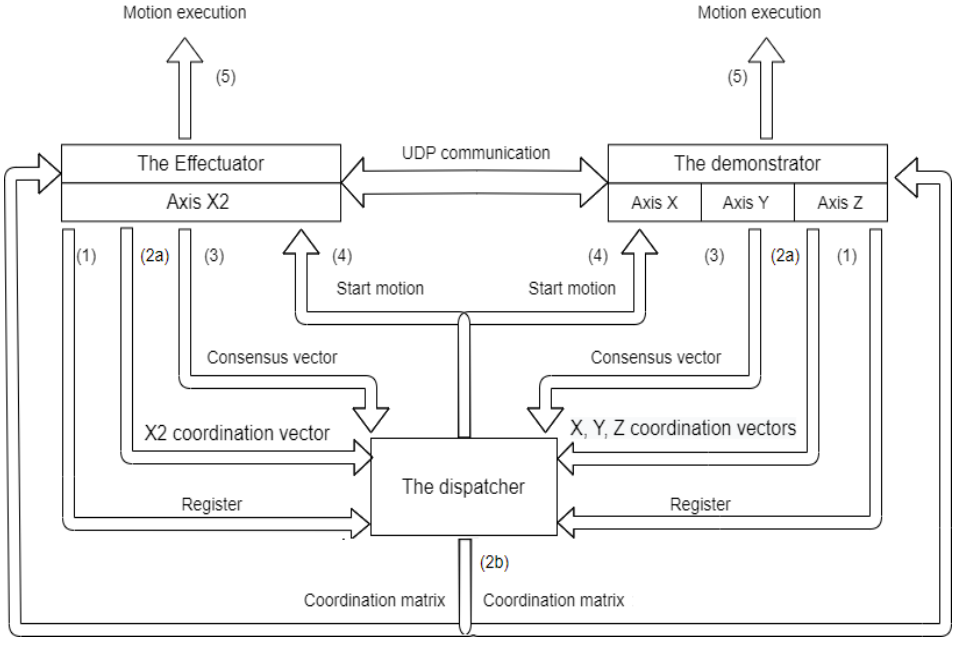
Conclusion and results





Conclusion and results

- A UDP communication network
- The Effectuator integration into the demonstrator axes
- The Effectuator motion control execution



The entire process of the new system [4]

Further work recommendations

- Driving a real DC motor by the Effectuator

DB42

Brushless DC Motor

DB42S03 Brushless DC Motor [6]

- Expanding the system with more embedded agents

References

- [1] https://www.isw.uni-stuttgart.de/forschung/engineering/devekos/
- [2] Caren Dripke, Daniel Schoebel, and Alexander Verl. Synchronization of motion-controlled axes with coordination vector and decentralized segment controller. 9 2020.
- [3] Ahmed Mahfouz. VHDL and embedded algorithms on an FPGA/NIOS CPU for distributed controller synchronization and motion control execution.Master's thesis, Institute for Control Engineering of Machine Tools and Manufacturing Units (ISW), Stuttgart,Germany, 7 2020
- [4] Pirmin Stanglmeier. Application case study for decentralized interpolation: Integration of commercial drive components and prototypical DEVEKOS axis control components. Master's thesis, Institute for Control Engineering of Machine Tools and Manufacturing Units (ISW), Stuttgart,Germany, 1 2020
- [5] Nisha Jiju. What's the difference between ip/tcp udp? https://www.colocationamerica.com/blog/tcp-ip-vs-udp, December 2018.
- [6] Nanotec. DB42 Brushless BB Motor. Technical report, info@nanotec.de,Munich,Germany, 1 2020.

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Thank you!

Ahmed Mahmoud Mohamed Aly Mahfouz

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